Ömer Can Atlı

Computer Engineer | Software Developer | AI & Backend

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Final-year Computer Engineering student at Gebze Technical University with a deep interest in backend systems and artificial intelligence. I build across the stack — .NET and Node.js backends, React Native and Flutter clients, and applied computer-vision and graph pipelines in Python. Creator and lead developer of two shipped products, and seven open-source projects published with the measurements that say where each one breaks.

WORK EXPERIENCE

Intern — Network Management Directorate July 2025 – August 2025

Türk Telekom, İstanbul

- Rotational internship across the Transmission, MPLS-IP, DSL, Mobile Core, Central Office and TTVPN units.
- Hands-on with xDSL/FTTH access technologies, MPLS, BNG, traffic monitoring and fault analysis on a national backbone network.

PRODUCTS

Shorties shorties.tr Creator & Lead Developer

- An exam-preparation ecosystem that turns a question bank into short vertical videos with AI: one shared architecture across three separate apps.
- Designed and built the .NET backend on a Clean Architecture split (API / Business / Core / DataAccess), the React Native (Expo) client and the Python video-generation engine, over MSSQL.
- YKS Shorties is live on the App Store and Google Play at version 2, with subscriptions through RevenueCat; KPSS and ALES are in development on the same stack.

C# · .NET · Clean Architecture · REST API · MSSQL · React Native (Expo) · Python · RevenueCat

respos resposapp.com Creator & Lead Developer

- A multi-tenant SaaS point-of-sale system for restaurants: one Node.js/Express backend serves many tenants, each with isolated data and self-service sign-up.
- Layered backend (routes → services → repositories) with tenant scoping in the service layer, JWT authentication and per-restaurant module access.
- Live table plan, kitchen display system, recipe-based stock tracking, shift and cash management, and discount authorisation with an audit log.

Node.js · Express · React · React Native (Expo) · JWT · Multi-tenancy

Mobile Fitness Tracker Team project

- Cross-platform fitness app building personalised meal and workout recommendations with AI; secure authentication, workout logging and data analytics.

Flutter · .NET · MSSQL

OPEN-SOURCE PROJECTS — github.com/Byomer1021

otonomarac · live demo Python · PyTorch · OpenCV · Gradio

Monocular driving perception and bird's-eye-view mapping

- Detection (YOLO), multi-object tracking (ByteTrack), monocular depth (Depth Anything V2), drivable-area segmentation, ground-plane homography and time-to-collision, built end to end over eight weeks.
- Deployed as a public Gradio demo on Hugging Face Spaces, running CPU-only on the free tier.

trafikisaret — Python · YOLO · Dataset construction

Turkish traffic-sign dataset and two-stage recogniser

- 717 bounding boxes labelled by hand over 787 frames of own dashcam footage in daylight, rain and night.
- Class-agnostic detector plus a 14-class crop classifier: 0.953 recall on a held-out test set; detector mAP50 0.515 overall — 0.608 in daylight, 0.361 in rain and at night.

smart-city-traffic-analysis — PySpark · GraphFrames · NetworkX · Pandas

38.3 million New York taxi trips modelled as a graph

- PageRank, Louvain communities and a node-removal simulation over 258 zones and 9,990 edges; full pipeline runs in 40.7 s.
- Removing the five most critical zones costs 34.1% of network flow; removing JFK Airport splits the graph from one connected component into sixteen.

plakatanima — Python · PyTorch · CTC · ONNX · TensorRT · C++

Turkish licence plate recognition, measured end to end

- Reads 18 of 34 plates correctly on a held-out test set fully automatically, and 22 with the corners marked by hand — the page leads with both, because only the first is what a camera would deliver. The four-plate gap is not resolved by this sample (McNemar p ~ 0.34).
- Synthetic generator calibrated against 272 hand-labelled real plates, a CTC recogniser fine-tuned on them, and a constrained beam search over the Turkish plate grammar written in C++.
- Exported to ONNX and TensorRT FP16 on a rented T4: 24.7x on YOLO and 5.4x on the recogniser, with all 157 crops decoding to identical plates.

payflow · **live demo** — Python · Recursive descent · Type systems · Pyodide

A domain-specific programming language for subscription billing

- Hand-written lexer, recursive-descent parser, type checker and tree-walking interpreter in pure Python — no parser generator, no third-party dependencies, 50 unit tests.
- Money carries its currency in the type, a domain-specific split operator rounds per step with banker's rounding, and paywall exhaustiveness is enforced by the grammar rather than checked afterwards.
- Runs live in the browser: the same five modules loaded into Pyodide, so a visitor sees the errors the real type checker raises rather than an imitation.

cse331-computer-organization — Verilog · MIPS assembly · Icarus Verilog · MARS

A MIPS program in assembly, and the processor built to run it

- Single-cycle MIPS32 processor in Verilog covering 36 integer instructions, and a gate-level datapath for the preceding assignment where a 32-bit adder is 32 chained full adders and set-less-than is sign XOR overflow.
- 370 self-checking assertions across the four assignments, run by one command. The MIPS text analyzer written in assembly runs on the processor built two assignments later, and the two agree on every count.

floorplan-parser — Python · OpenCV · scikit-image · NetworkX

Floor plans to walls, doors, windows and a room graph, no learned models

- Two-scale morphology, skeleton and probabilistic Hough lines, a quarter-arc door template and a distance-transform watershed turn a raster plan into walls, openings, rooms and a room-door graph; no neural network anywhere.
- Thresholds chosen on 62 development plans and the held-out test split measured once: wall IoU 0.913, door F1 0.904, window F1 0.830, room F1 0.770 under strict one-to-one matching.

EDUCATION

Gebze Technical University — Expected 2027 · Kocaeli

B.Sc. Computer Engineering

Şehit Mustafa Serin Science High School — Graduated 2019 · Balıkesir

Science track

TECHNICAL SKILLS

Languages Java, C++, C, C#, Python, JavaScript, SQL, HTML/CSS

Frameworks & Libraries	.NET, Spring Boot, Django, Flutter, React Native, Scikit-learn, TensorFlow (basic), NumPy, Pandas
Tools & Platforms	Git, Docker, Jira, Maven, Firebase, MSSQL, MySQL, SQL Server
Core Concepts	Object-Oriented Programming, Data Structures, Machine Learning, Deep Learning

LANGUAGES

Turkish — native · English — Advanced (C1)

VOLUNTEER WORK & EXTRACURRICULARS

C developer, "Besleme Kahramanları" · Member, GTÜ Bilgisayar Topluluğu · AFAD disaster response volunteer · Member, GTU Search and Rescue Club (GETAK)